XING HAN

Homepage | Google Scholar | LinkedIn | xhan56@jh.edu | +1 (512)-987-5026

Research Vision

I design multimodal foundation models and agents capable of human-expert reasoning in high-stakes environments. My research bridges core machine learning with real-world deployment such as healthcare. My vision is to build trustworthy, self-improving AI systems that seamlessly integrate into workflows to improve outcomes, topics include:

- **Multimodal Foundation Models:** multimodal fusion, cross-modal interactions, medical foundation models
- **Self-Evolving Agentic System:** self-play methods, memory and knowledge evolution, counterfactual simulation
- **Trustworthy & Enduring Deployment:** uncertainty quantification, continuous monitoring, interpretability
- **Foundations of Machine Learning:** time-series sequence models, ensemble models, neural scaling laws

Education

Postdoctoral Fellow in Computer Science — Sept. 2023 – now
Johns Hopkins University Data Science and AI Institute — Baltimore, MD & Mountain View, CA, USA
Advisor: Suchi Saria Co-advisor: Gregory D. Hager

M.Sc. - Ph.D. in Electrical and Computer Engineering — August 2023
The University of Texas at Austin — Austin, TX, USA
Advisor: Joydeep Ghosh Committee: Aryan Mokhtari, Atlas Wang, Mingyuan Zhou, Nhat Ho
Dissertation: Reliable Predictions for Structured and Corrupted Data

B.Eng. in Electrical and Electronics Engineering — July 2017
The University of Edinburgh — Edinburgh, UK
Advisor: Jiabin Jia GPA: 3.92 (graduated with first-class honor)
University of Edinburgh "2+2" Scholarship in 2015 & 2016

Publications

XH denotes the author; * denotes equal contribution; ✉ denotes corresponding author; ★ marks selected work.

Published Works

[1] **On the Invariance and Generality of Neural Scaling Laws** (23 pages) ★
XH, Z. Liu, S. Saria, and P. Liang
Neural Information Processing Systems (NeurIPS) 2026

[2] **FLAME: Adaptive Mixture-of-Experts for Continual Multimodal Multi-Task Learning** (37 pages) ★
XH, S. Chaudhari, T. Ranade, R. Chellappa, and S. Saria
Neural Information Processing Systems (NeurIPS) 2026

[3] **Massively Multimodal Foundation Models: A Frameworks for Capturing Interactions with Specialized Mixture-of-Experts** (28 pages) ★
Top ~10% of accepted papers by reviewer rating
XH, H. Chung, J. Ghosh, P. Liang, and S. Saria
International Conference on Learning Representations (ICLR) 2026

[4] **QoQ-Med3: Robust Multimodal Clinical Analysis Foundation Model with Reasoning** (23 pages) ★
W. Dai, J. She, J. Cheong, **XH**, C. Harris, H. Wei, F. Vahedifard, S. Saria, R. Stevens, and P. Liang
NPJ Digital Medicine 2026

[5] **WATCH: Adaptive Monitoring for AI Deployments via Weighted-Conformal Martingales** (30 pages) ★
D. Prinster ✉, **XH**✉, A. Liu, and S. Saria
International Conference on Machine Learning (ICML) 2025

[6] **Between Linear and Sinusoidal: Rethinking the Time Encoder in Dynamic Graph Learning** (25 pages)
H. Chung, S. Chaudhari, **XH**, Y. Wald, S. Saria, and J. Ghosh
Transactions on Machine Learning Research (TMLR) 2025

[7] **SSL-DA: Semi- and Self-Supervised Learning with Dual Attention for Echocardiogram Segmentation** (14 pages)
L. Lv, **XH**, Z. Sun, Z. Li, X. Wang, T. Jiang, Y. Liu, T. Li, J. Xu, L. You, G. Yao, F. Sun, and J. Xing
Journal of Imaging Informatics in Medicine 2025

[8] **FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion** (47 pages) ★
Over 185 citations as of October 2026
XH, H. Nguyen, C. Harris, N. Ho, and S. Saria
Neural Information Processing Systems (NeurIPS) 2024

[9] **Novel Node Category Detection Under Subpopulation Shift** (24 pages)
H. Chung, S. Chaudhari, Y. Wald, **XH**, and J. Ghosh
European Conference on Machine Learning and Data Mining (ECML-PKDD) 2024

[10] **Designing Robust Transformers using Robust Kernel Density Estimation** (23 pages) ★
XH, T. Ren, T. Nguyen, K. Nguyen, J. Ghosh, and N. Ho
Neural Information Processing Systems (NeurIPS) 2023

[11] **Efficient Forecasting of Large Scale Hierarchical Time Series via Multilevel Clustering** (17 pages)
XH, T. Ren, J. Hu, J. Ghosh, and N. Ho
International Conference on Time Series and Forecasting (ITISE) 2023

[12] **A Novel Control-Variates Approach for Performative Gradient-Based Learners with Missing Data** (8 pages)
XH, J. Hu, and J. Ghosh
International Joint Conference on Neural Networks (IJCNN) 2023

[13] **Architecture Agnostic Federated Learning for Neural Networks** (11 pages) ★
D. Makhija, **XH**, N. Ho, and J. Ghosh
International Conference on Machine Learning (ICML) 2022

[14] **Dynamic Combination of Heterogeneous Models for Hierarchical Time Series** (10 pages)
XH, J. Hu, and J. Ghosh
International Conference on Data Mining Workshops (ICDMW) 2022

[15] **Simultaneously Reconciled Quantile Forecasting of Hierarchically Related Time Series** (17 pages) ★
XH, S. Dasgupta, and J. Ghosh
Artificial Intelligence and Statistics (AISTATS) 2021

[16] **Split Localized Conformal Prediction** (21 pages)
XH, Z. Tang, J. Ghosh, and Q. Liu
ICML Workshop on Distribution-Free Uncertainty Quantification 2021

[17] **Multi-Pair Text Style Transfer for Unbalanced Data via Task-Adaptive Meta-Learning** (8 pages)
XH and J. Lundin
ACL MetaNLP Workshop 2021

[18] **Model-Agnostic Explanations using Minimal Forcing Subsets** (8 pages)
XH and J. Ghosh
International Joint Conference on Neural Networks (IJCNN) 2021

[19] **Certified Monotonic Neural Networks** (17 pages) ★
Spotlight presentation: 280/9454 ≈ 2.96%; over 202 citations as of October 2026
X. Liu, **XH**, N. Zhang, and Q. Liu
Neural Information Processing Systems (NeurIPS) 2020

[20] **Transparent Interpretation with Knockout** (8 pages)
XH, Y. Feng, N. Zhang, and Q. Liu
ICML Workshop on Human Interpretability in Machine Learning (WHI) [Spotlight] 2020

[21] **Sensing Personality to Predict Job Performance** (10 pages)
S. Lin, S.M. Mattingly, **XH**, P. Audia, et al.
CHI Workshop on The Future of Work 2019

[22] **Effects of Integrated Intent Recognition and Communication on Human-Robot Collaboration** (6 pages)
A. Gutierrez, M.L. Chang, **XH**, and K.C. Chang
International Conference on Intelligent Robots and Systems (IROS) 2018

Preprints

[23] **Counterfactual Self-Evolving Agents for Evidence-Grounded Reasoning** (38 pages) ★
XH, Y. Wang, C. Chen, W. Dai, G. Gudur, S. Li, H. Chung, G. Hager, J. Ghosh, P. Liang, and S. Saria
In submission to ICLR 2027 2026

[24] **ADMIT: Admissible Diffusion for Multimodal Interventional Trajectories**
XH, S. Chaudhari, J. Shao, P. Liang, and S. Saria
Preprint 2026

[25] **MILM: Large Language Models for Multimodal Irregular Time Series with Informative Sampling** (29 pages)
H. Chung, S. Li, Y. Wald, **XH**, S. Saria, and J. Ghosh
In submission to ICLR 2027 2026

[26] **RSIMed: Scaling Medical Foundation Models beyond Limited Clinical Data via Retrieval-Grounded Self-Evolution**
W. Dai, Q. Liu, G. Qin, **XH**, T. Naumann, S. Zhang, and P. Liang
In submission to ICLR 2027 2026

[27] **Recursive Reward Improvement via Adversarial Meta-Prompt Evolution** (38 pages)
W. Dai, Q. Liu, G. Qin, **XH**, T. Naumann, P. Liang, and S. Zhang
In submission to ICLR 2027 2026

[28] **MINT: Multimodal Instruction Tuning with Multimodal Interaction Grouping** (26 pages)
X. Shan*, Q. Cao*, **XH***, H. Yu*, and P. Liang
Preprint 2025

[29] **Quadratic Gating Functions in Mixture of Experts: A Statistical Insight** (53 pages)
P. Akbarian*, H. Nguyen*, **XH***, and N. Ho
In submission to SIAM Journal on Mathematics of Data Science 2024

[30] **On Expert Estimation in Hierarchical Mixture of Experts: Beyond Softmax Gating Functions** (68 pages)
H. Nguyen*, **XH***, C. Harris, N. Ho, and S. Saria
In submission to JMLR 2024

[31] **Achieving Fairness Across Local and Global Models in Federated Learning** (15 pages)
D. Makhija, **XH**, J. Ghosh, and Y. Kim
Preprint 2024

Patents

[1] **Machine-Learning Based Generation of Text Style Variations for Digital Content Items**
J. Lundin, O. Schoppe, **XH**, M. Sollami, B. Lonsdorf, A. Ross, D. Woodward, and S. Rohde
US Patent 2022/0245322 A1 2022

Grants & Protocols

Key Personnel, ARPA-H Emerging Health Innovators Program *In progress*

· Oversee the core technical component (causal reasoning alignment and verification) of "Causally-Grounded Alignment for Clinical Generative Artificial Intelligence Tools". PI: Dr. Adarsh Subbaswamy (University of Maryland).

Key Personnel, Patient-Centered Outcomes Research Institute (PCORI) Grant *In progress*

· Lead and oversee the core technical component (multimodal foundation model development and post-deployment monitoring) of "Improving Methods for Conducting Patient-Centered Comparative Clinical Effectiveness Research"; drafted the letter of intent and grant proposals for submission. PI: Dr. Suzanne Simkovich (MedStar).

Grant Drafter, Amazon–JHU Health Tech Research Grant *In progress*

· Drafting a proposal on formal safety constraints, subgroup risk control, and monitoring of personalized medical AI across model updates. PI: Dr. Suchi Saria (Johns Hopkins University).

Primary Contributor, Gordon and Betty Moore Foundation Grant, ∼ **$500K** in total *2023 – now*

· Developed multimodal clinical foundation models and post-deployment continuous monitoring methods that served as core technical contributions. PI: Dr. Suchi Saria (Johns Hopkins University).

Key Personnel, Intuit AI Research Grants, **$300K** in total *2020, 2021, 2022*

· Drafted the proposals and budget planning documents for **three separate annual grants** of $100,000 each, submitted in 2020, 2021, and 2022: **all three awarded**. Developed a series of models in time-series forecasting and clustering, deployed in industry-level financial forecasting pipelines. PI: Dr. Joydeep Ghosh (UT-Austin).

Protocol Drafter and Submitter, Johns Hopkins Medicine IRB *2024 – now*

· Drafted, submitted, and maintained annual renewals for the following IRB protocols:
  – **IRB00464029**, "Leveraging Multimodal Patient Data for Enhanced Clinical Task Performance"
  – **CR00054024**, "Development and Validation of Robust Models for Adverse Event Risk Prediction"

Academic Employments

Postdoctoral Fellow, Johns Hopkins University Baltimore, MD & Mountain View, CA
Supervisors: Prof. Suchi Saria, Prof. Gregory D. Hager *09/01/2023 – now*

· Designing multimodal transformers and self-evolving agents for medical applications; advising graduate and undergraduate students; collaborating with researchers and clinicians on grants; and drafting IRB protocols

Visiting Researcher, Massachusetts Institute of Technology — Cambridge, MA
EECS and Media Lab, Supervisor: Prof. Paul Pu Liang — *02/2025 – 07/2025*

- Worked on emerging interaction-based multimodal foundation models, medical agents, and neural scaling laws for these models, and on large-scale fine-tuning of multimodal clinical LLMs

Graduate Research Assistant, The University of Texas at Austin — *06/01/2019 – 08/14/2023*
Intelligent Data Exploration and Analysis Lab, Department of Electrical and Computer Engineering
Supervisor: Prof. Joydeep Ghosh — Austin, TX

- Research topics include time-series forecasting, clustering, uncertainty quantification, federated learning, robustness and interpretability of machine learning

Data Science Group, Department of Statistics and Data Sciences
Supervisor: Prof. Nhat Ho — Austin, TX

- Research topics include optimal transport, Bayesian deep learning, model fusion/alignment, mixture of experts, and robustness of Transformer models

Statistical Learning and AI Group, Department of Computer Science
Supervisor: Prof. Qiang Liu — Austin, TX

- Research topics include the robustness and interpretability of deep neural networks, conformal prediction, generative models, and predictive modeling of mobile sensor data

Industry Employments

Google — Sunnyvale, CA
Research Scientist Intern (Host: Dr. Xu Gao) — *05/31/2022 – 09/02/2022*

- Enhanced the workload prediction accuracy for Google Cloud machines by 20% compared to baseline
- Utilized explanation tools (Shapley value) to understand the root causes of machine downtime

Intuit — Mountain View, CA
Data Scientist Intern (Mentor: Dr. Jing Hu) — *06/15/2021 – 09/03/2021*

- Improved cash-flow forecasting accuracy by 10% and deployed models to applications
- Decreased computation time by 45% through the development of a parallelized program

Salesforce — San Francisco, CA
Research Intern (Mentor: Dr. Jessica Lundin) — *06/01/2020 – 08/21/2020*

- Implemented 3 different state-of-the-art large language models (LLM) for text style transfer
- Leveraged meta-learning to fine-tune pre-trained LLM on a small and unbalanced dataset

CognitiveScale — Austin, TX
Applied Scientist Intern (Mentor: Dr. Suyog Jain) — *05/29/2018 – 08/03/2018*

- Reproduced and improved 5 different session-based recommendation algorithms
- Led a team project of 4 interns to research "Ethical AI" and pitched a business idea to company

Community Services

Conference Reviewer/PC Member

International Conference on Learning Representations (ICLR) [AI/ML]	2020 – now
Neural Information Processing Systems (NeurIPS) [AI/ML]	2020 – now
International Conference on Machine Learning (ICML) [AI/ML]	2021 – now
Artificial Intelligence and Statistics (AISTATS) [AI/ML, Statistics]	2021 – now

International Joint Conference on Neural Networks (IJCNN) [AI/ML] 2023 – now
Machine Learning for Health (ML4H) [Health] 2023 – now
International Conference on Acoustics, Speech and Signal Processing (ICASSP) [Signal Processing] 2023 – now
Machine Learning for Healthcare (MLHC) [Health] 2024 – now
SIAM Conference on Data Mining (SDM) [Data Mining] 2026 – now
Computer Vision and Pattern Recognition (CVPR) [Computer Vision] 2025 – now
European Conference on Computer Vision (ECCV) [Computer Vision] 2026 – now

Journal Reviewer

Pattern Recognition [AI/ML] 2019 – now
Journal of Open Source Software [Multidisciplinary] 2024 – now
IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI) [AI/ML] 2025 – now
Nature Biomedical Engineering [Biomedical] 2025 – now
Medical Image Analysis [Biomedical] 2026 – now

Other Academic Services

Lead Organizer, ICLR 2027 Workshop on Human-Centered AI for Health 2027
Session Chair in Trustworthy ML, SDM 2026 2026
Technical Program Committee, WellComp Workshop at UbiComp/ISWC 2026
Reading Group Organizer, Multimodal Learning in Health at JHU 2024 – 2025
Volunteer Assistant, UT-Austin ECE Graduate Admission Committee 2021 – 2023
Volunteer and Moderator, ICML Workshops 2020, 2021, 2023
Third-year Class Representative, University of Edinburgh 2015 – 2016

Selected Honours and Awards

ICML Gold Reviewer Award 2026
ICLR Travel Grant, $2000 2026
NeurIPS Scholar Award, $2000 2023 – 2024
AHLI CHIL Doctoral Symposium Spotlight 2023
NeurIPS Spotlight Presentation 2020
Excellent Teaching Assistant Award, $1000 2018
Outstanding Graduates of Beijing City (top 3%) 2017
University of Edinburgh 2 + 2 Scholarship, £2500 per year 2015 – 2016
Comprehensive Scholarship, ¥5000 2014 – 2015
China National Scholarship, ¥8000 2013 – 2014

Mentorship & Teaching

Mentorship

Mentor, ML4H Author & Reviewer & Career Mentorship Program 2024 – now
– Mentees: Yang Li (William & Mary), Takeshi Hara (St. Marianna University), Ching-Tang Wang (Tohoku University)
Shravan Chaudhari, Ph.D. Student in Computer Science JHU, 2023 –
Topic: Multimodal and multitask pretraining, continual learning → NeurIPS 2026
Carl Harris, Ph.D. Student in Biomedical Engineering JHU, 2023 – 2025
Topic: Multimodal irregular temporal sequence modeling, electronic health records → NeurIPS 2024
Chen Chen, Master Student in Computer Science JHU, 2026 –
Topic: Retrieval-augmented generation, counterfactual reasoning → ICLR 2027, under review
Yuxin Wang, Master Student in Applied Mathematics JHU, 2026 –
Topic: LLM fine-tuning, counterfactual reasoning, reinforcement learning → ICLR 2027, under review

Jiarui Shao, Master Student in Computer Science — JHU, 2026 –
Topic: Multimodal foundation model, diffusion model, multimodal interactions

Julia Lee, Undergraduate Student in NeuroScience — JHU, 2026 –
Topic: Multimodal foundation model, healthcare applications

Chan Um, Master Student in Biomedical Engineering — JHU, 2026 –
Topic: Continuous monitoring of deployed multimodal models

Tanvi Ranade, Undergraduate Student in Computer Science — JHU, 2024 – 2026
Topic: Multimodal and multitask pretraining, continual learning → NeurIPS 2026

Hsing-Huan Chung, Ph.D. Student in ECE — UT-Austin, 2023 – 2026 (now with Uber)
Topic: Time-series model, graph neural networks, domain adaptation → TMLR, ECML-PKDD 2024

Disha Makhija, Ph.D. Student in ECE — UT-Austin, 2023 – 2024 (now with Amazon)
Topic: Federated learning → ICML 2022

Kurtis E. David, Master Student in Computer Science — UT-Austin, 2019 – 2020 (now with DeepMind)
Topic: Certified robustness of neural networks

Teaching

Lecturer, self-developed short course on Multimodal Learning in Healthcare [Certificate] — JHU, 2025
Teaching Assistant, EE 381V, Fair Transparent Machine Learning — UT-Austin, 2022
Teaching Assistant, EE 422C, Software Design & Implementation — UT-Austin, 2018
Lab Instructor, ELEE09024, Digital System Design 3 — University of Edinburgh, 2016

Invited Talks & Presentations

Guiding Mixture-of-Experts in Multimodal Learning: Temporal Dynamics and Robust Fusion — *EcoSta*, 2025
FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion — *MIT Media Lab*, 2025
FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion — *Columbia University*, 2024
Reliable Predictions for Structured and Corrupted Data — *Fudan University*, 2023
Hierarchical Time Series Forecasting — *IIF Workshop on Forecast Reconciliation*, 2023
Reliable Prediction of Treatment Effect Through Time — *Intuit AI*, 2023
Time Series Reconciliation with Dynamically Changing Structures — *Google Cloud AI*, 2022
Mixture-of-Experts for Probabilistic Forecasts of Aggregated Time Series — *Intuit AI*, 2021
Certified Monotonic Neural Networks — *Salesforce Research*, 2020
Transparent Interpretation with Knockout — *Intuit AI*, 2019

References

Suchi Saria, John C. Malone Associate Professor (Postdoc Advisor) — ssaria@cs.jhu.edu
Department of Computer Science, Johns Hopkins University — Baltimore, MD
CEO and CSO, Bayesian Health — New York, NY

Joydeep Ghosh, Schlumberger Centennial Chair Professor (PhD Advisor) — jghosh@utexas.edu
Department of Electrical and Computer Engineering, UT-Austin — Austin, TX

Paul Pu Liang, Assistant Professor — ppliang@mit.edu
Media Lab and EECS, MIT — Cambridge, MA

Gregory D. Hager, Mandell Bellmore Professor (Postdoc Co-Advisor) — hager@jhu.edu
Department of Computer Science, Johns Hopkins University — Baltimore, MD

Nhat Ho, Associate Professor — minhnhat@utexas.edu
Department of Statistics and Data Sciences, UT-Austin — Austin, TX